

In [1]:

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    "Month": ["Jan", "Feb", "Mar", "Apr", "May", "Jun"],
    "Sales": [120, 150, 170, 160, 200, 230]
}

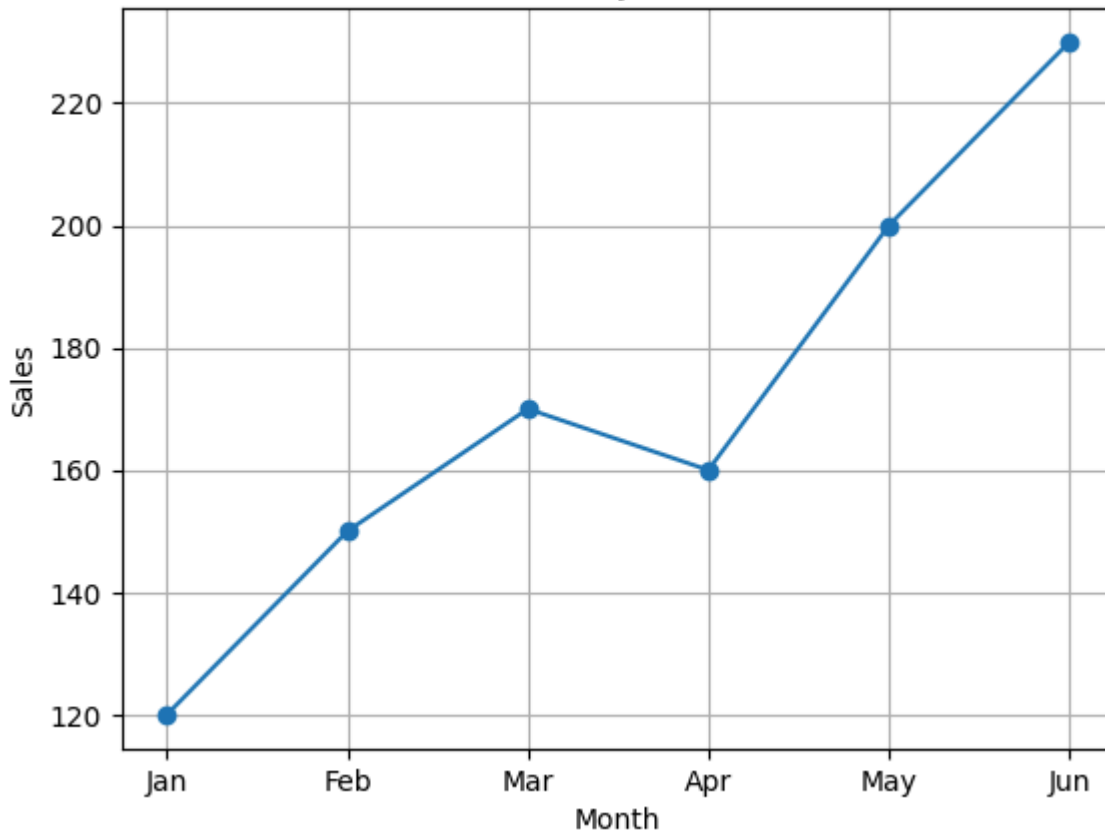
df = pd.DataFrame(data)

plt.plot(df["Month"], df["Sales"], marker="o")

plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")

plt.grid()
plt.show()
```

Monthly Sales



In [2]:

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    "Student": ["Amit", "Sara", "John", "Neha", "Ali"],
    "Marks": [78, 92, 65, 88, 75]
}

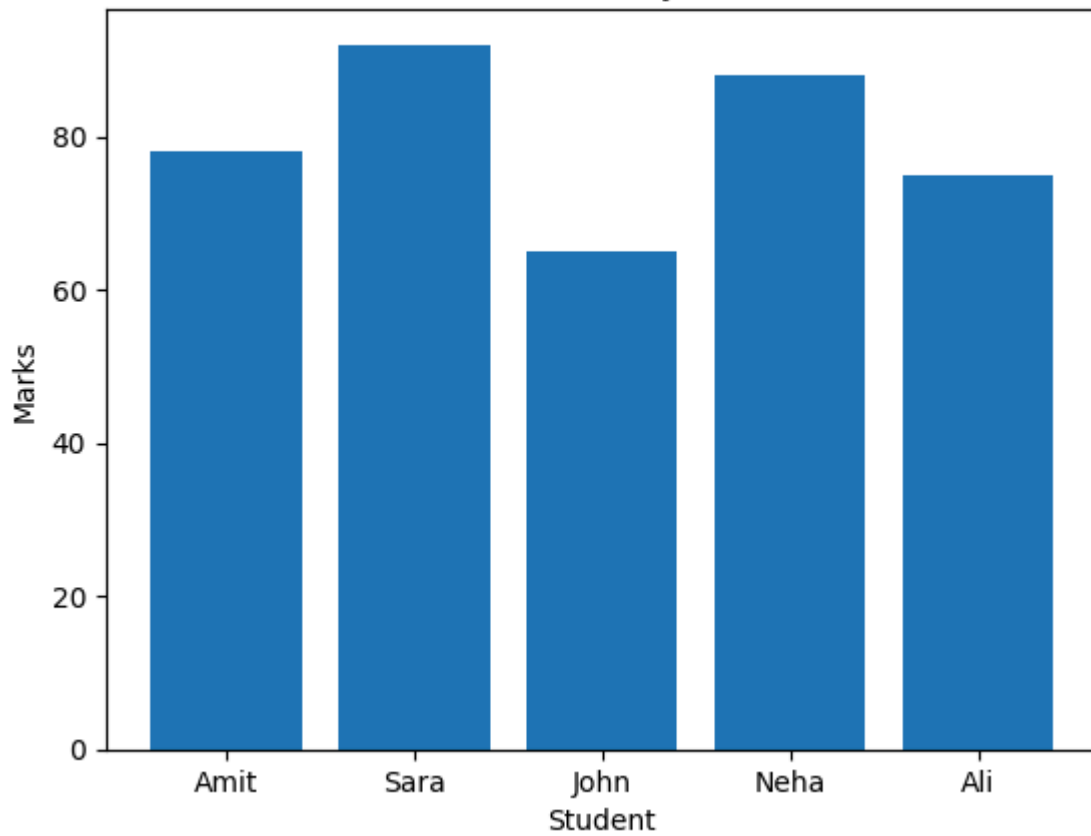
df = pd.DataFrame(data)

plt.bar(df["Student"], df["Marks"])

plt.title("Marks Obtained by Students")
plt.xlabel("Student")
plt.ylabel("Marks")

plt.show()
```

Marks Obtained by Students



In [3]:

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    "Language": ["Python", "Java", "C++", "JavaScript", "R"],
    "Students": [120, 90, 70, 100, 40]
}

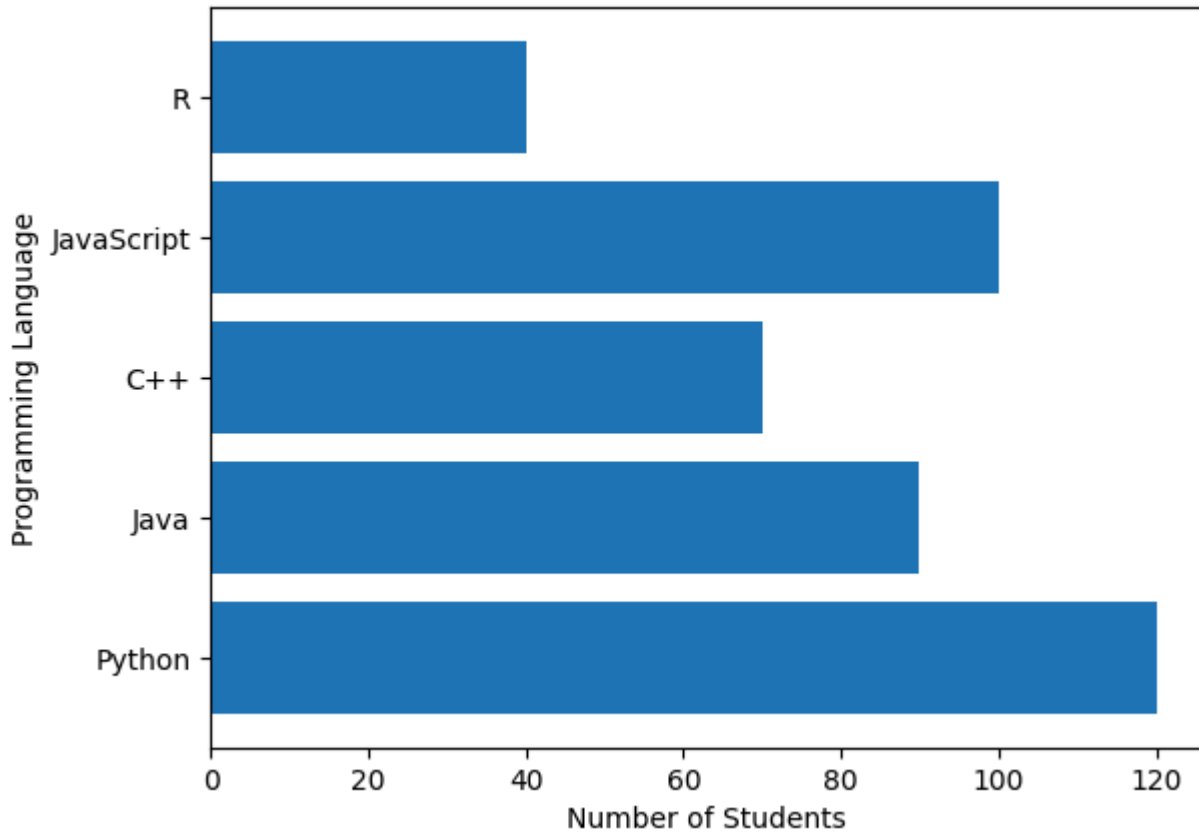
df = pd.DataFrame(data)

plt.barh(df["Language"], df["Students"])

plt.title("Programming Language Popularity")
plt.xlabel("Number of Students")
plt.ylabel("Programming Language")

plt.show()
```

Programming Language Popularity



In [4]:

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    "Department": ["Marketing", "IT", "HR", "Operations", "Research"],
    "Budget": [30, 25, 15, 20, 10]
}

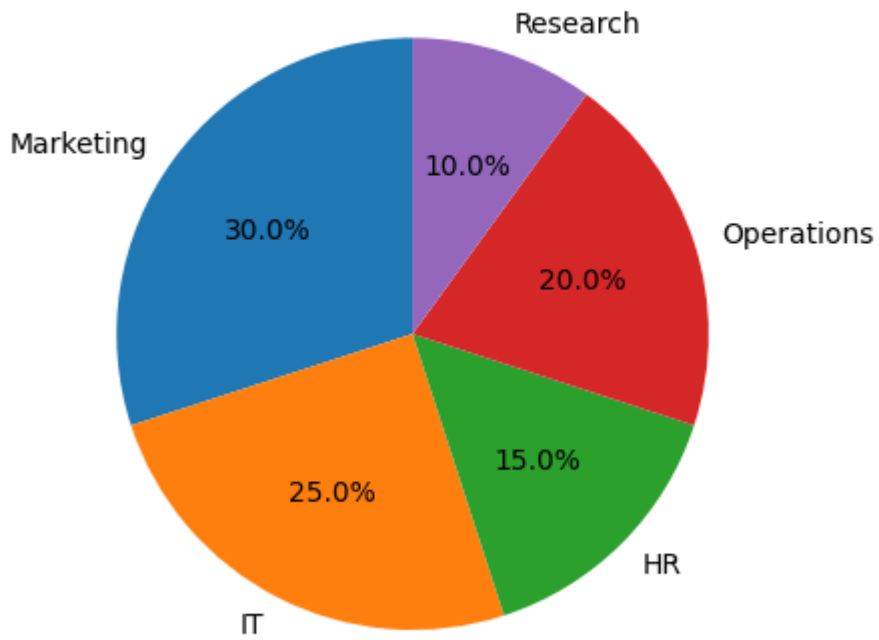
df = pd.DataFrame(data)

plt.pie(
    df["Budget"],
    labels=df["Department"],
    autopct="%1.1f%%",
    startangle=90
)

plt.title("Company Budget Distribution")

plt.show()
```

Company Budget Distribution



In [5]:

```
import matplotlib.pyplot as plt

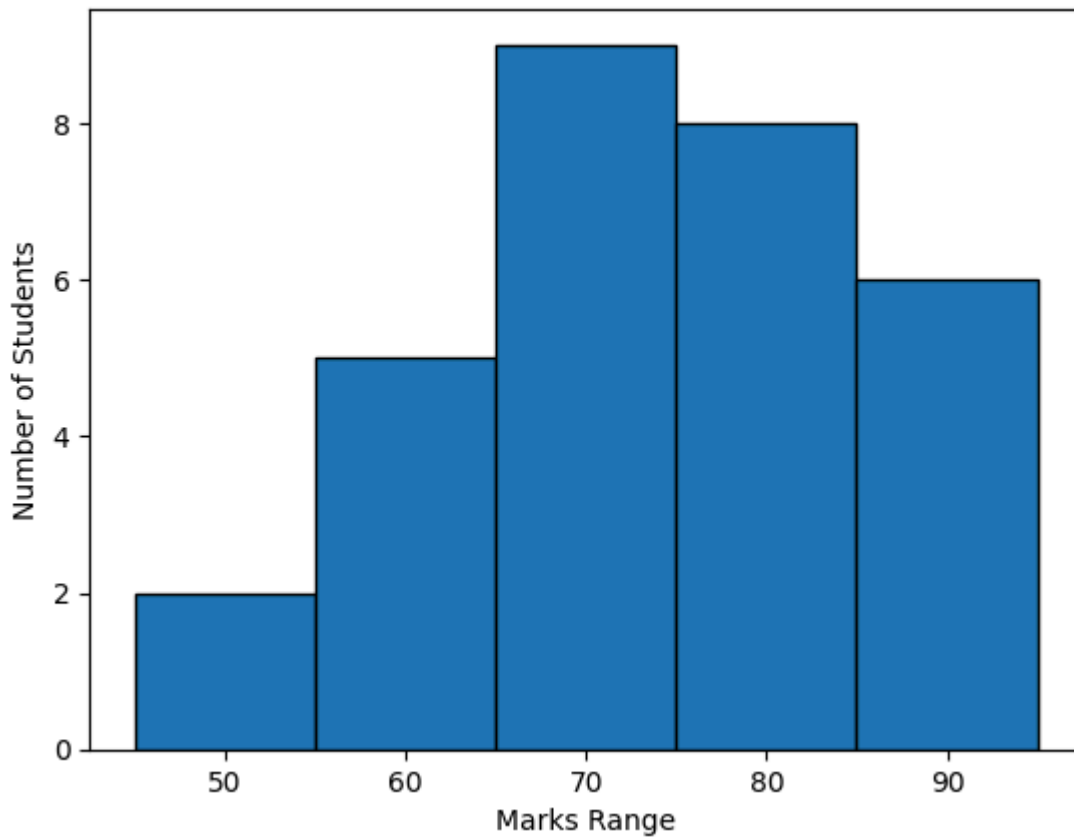
marks = [
    45, 52, 56, 58, 61,
    62, 63, 65, 67, 68,
    69, 70, 71, 72, 73,
    74, 75, 76, 77, 78,
    79, 80, 82, 83, 85,
    86, 88, 90, 92, 95
]

plt.hist(marks, bins=5, edgecolor="black")

plt.title("Distribution of Student Marks")
plt.xlabel("Marks Range")
plt.ylabel("Number of Students")

plt.show()
```

Distribution of Student Marks



In [6]:

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    "Study_Hours": [1, 2, 3, 4, 5, 6, 7, 8, 9],
    "Marks": [45, 50, 58, 65, 70, 78, 82, 88, 92]
}

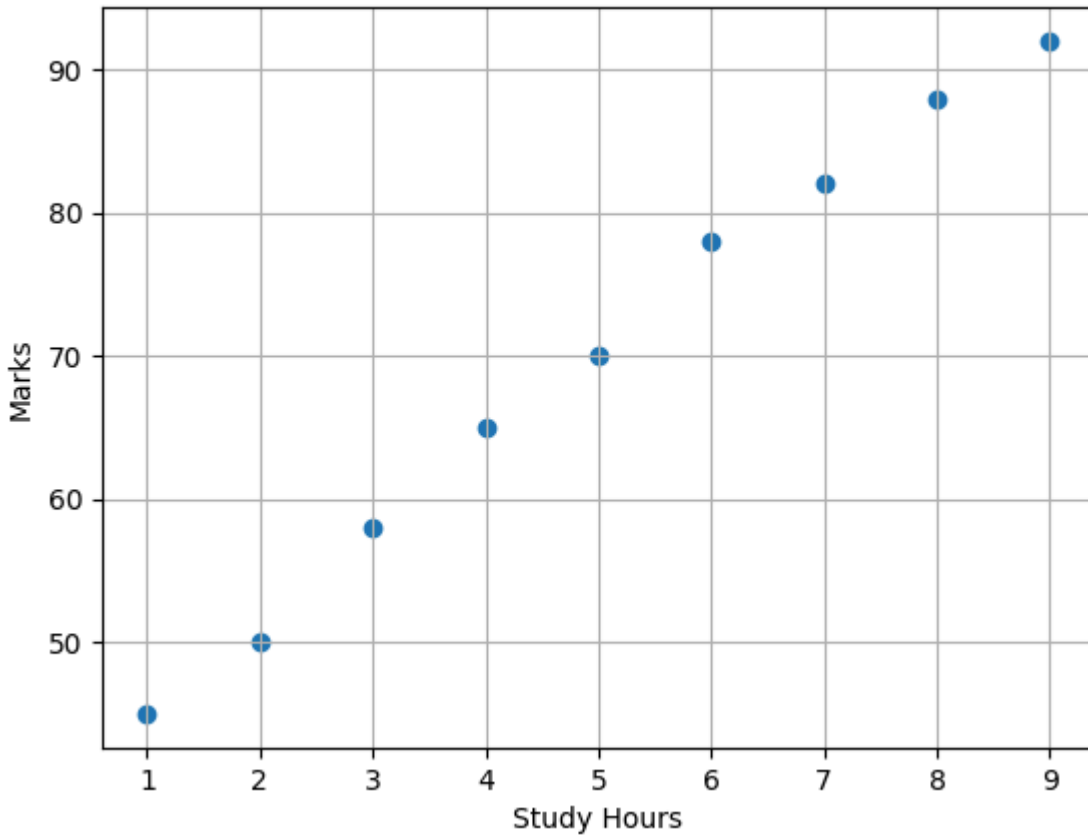
df = pd.DataFrame(data)

plt.scatter(df["Study_Hours"], df["Marks"])

plt.title("Study Hours vs Marks")
plt.xlabel("Study Hours")
plt.ylabel("Marks")

plt.grid()
plt.show()
```

Study Hours vs Marks



In [7]:

```
import matplotlib.pyplot as plt

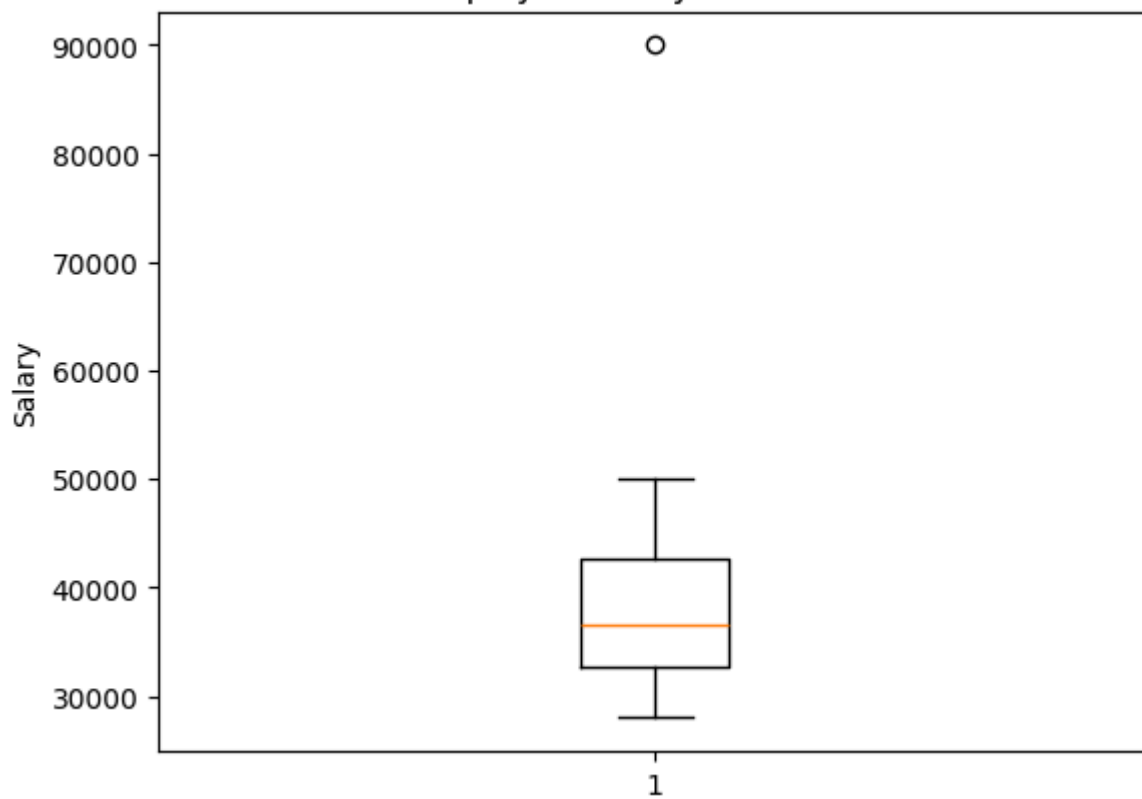
salaries = [
    28000, 30000, 31000, 32000,
    33000, 34000, 35000, 36000,
    37000, 38000, 40000, 42000,
    45000, 47000, 50000,
    90000
]

plt.boxplot(salaries)

plt.title("Employee Salary Distribution")
plt.ylabel("Salary")

plt.show()
```

Employee Salary Distribution



In [8]:

```
import matplotlib.pyplot as plt

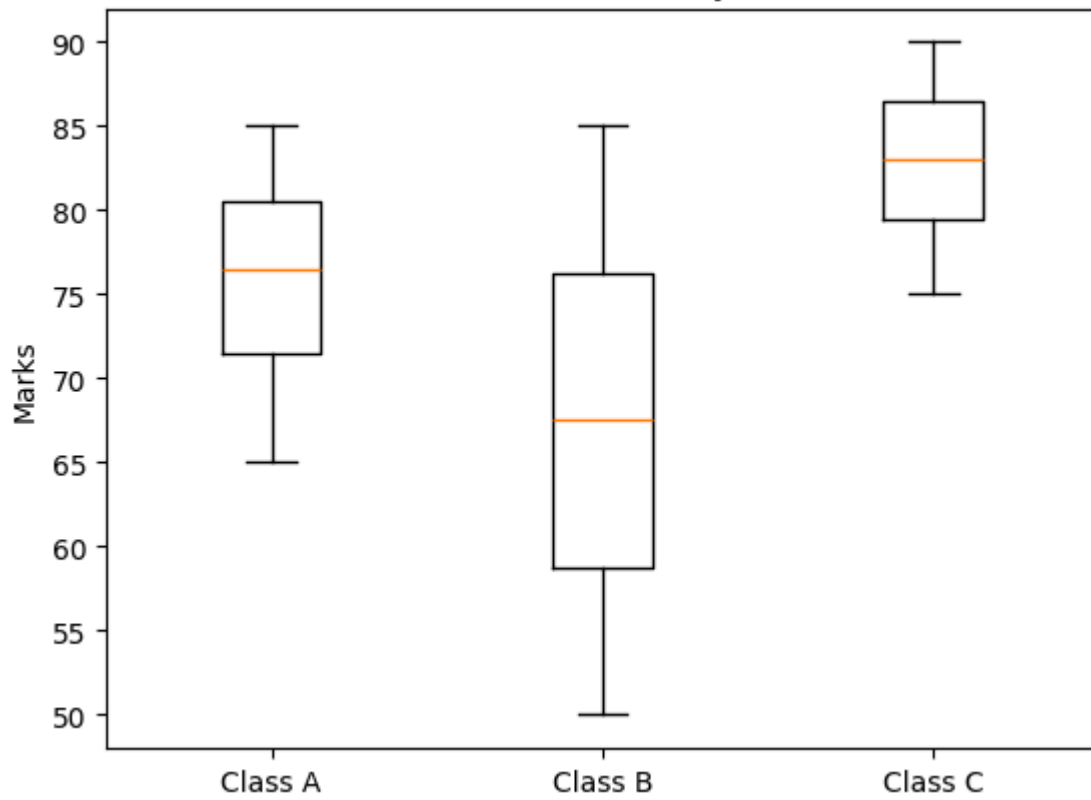
class_A = [65, 70, 72, 75, 78, 80, 82, 85]
class_B = [50, 55, 60, 65, 70, 75, 80, 85]
class_C = [75, 78, 80, 82, 84, 86, 88, 90]

plt.boxplot(
    [class_A, class_B, class_C],
    tick_labels=["Class A", "Class B", "Class C"]
)

plt.title("Marks Distribution by Class")
plt.ylabel("Marks")

plt.show()
```

Marks Distribution by Class



In [9]:

```
import matplotlib.pyplot as plt

months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
visitors = [1000, 1400, 1800, 1600, 2200, 2700]

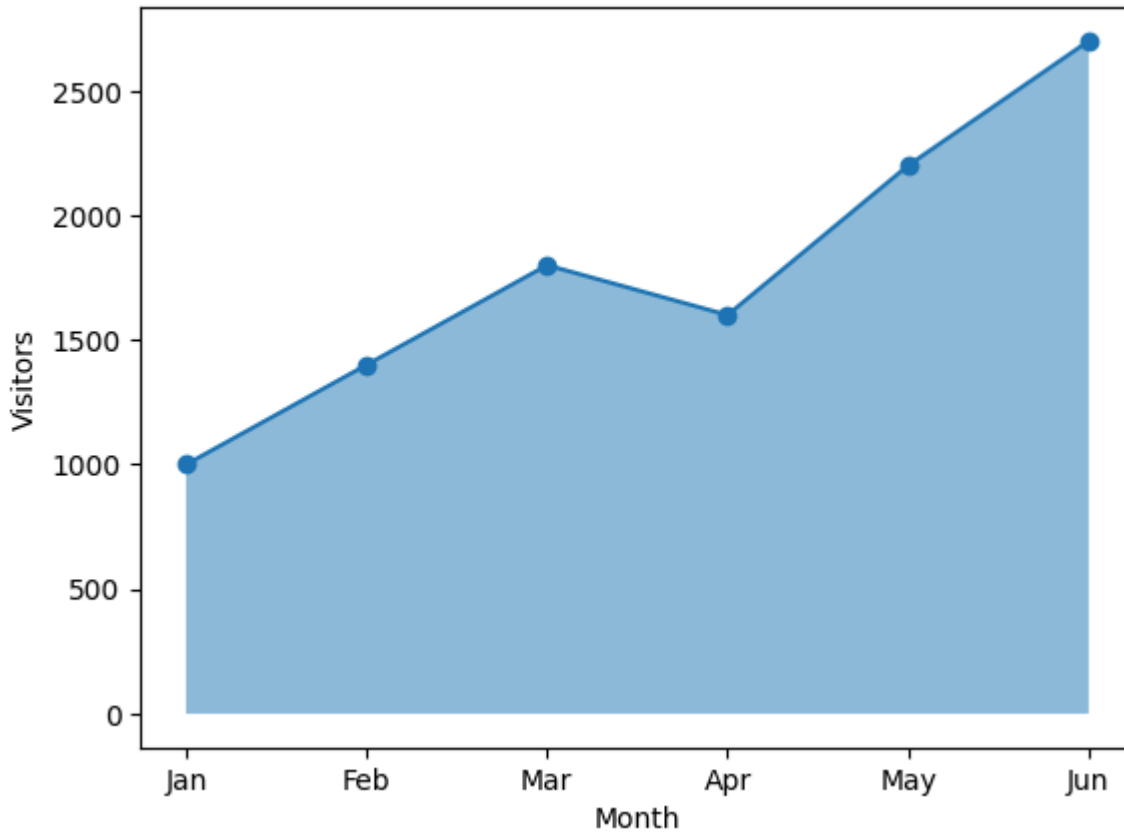
plt.fill_between(months, visitors, alpha=0.5)

plt.plot(months, visitors, marker="o")

plt.title("Website Visitors")
plt.xlabel("Month")
plt.ylabel("Visitors")

plt.show()
```

Website Visitors



In [10]:

```
import matplotlib.pyplot as plt
import numpy as np

quarters = ["Q1", "Q2", "Q3", "Q4"]

product_A = [120, 150, 170, 200]
product_B = [100, 130, 160, 190]

x = np.arange(len(quarters))
width = 0.35

plt.bar(x - width/2, product_A, width, label="Product A")
plt.bar(x + width/2, product_B, width, label="Product B")

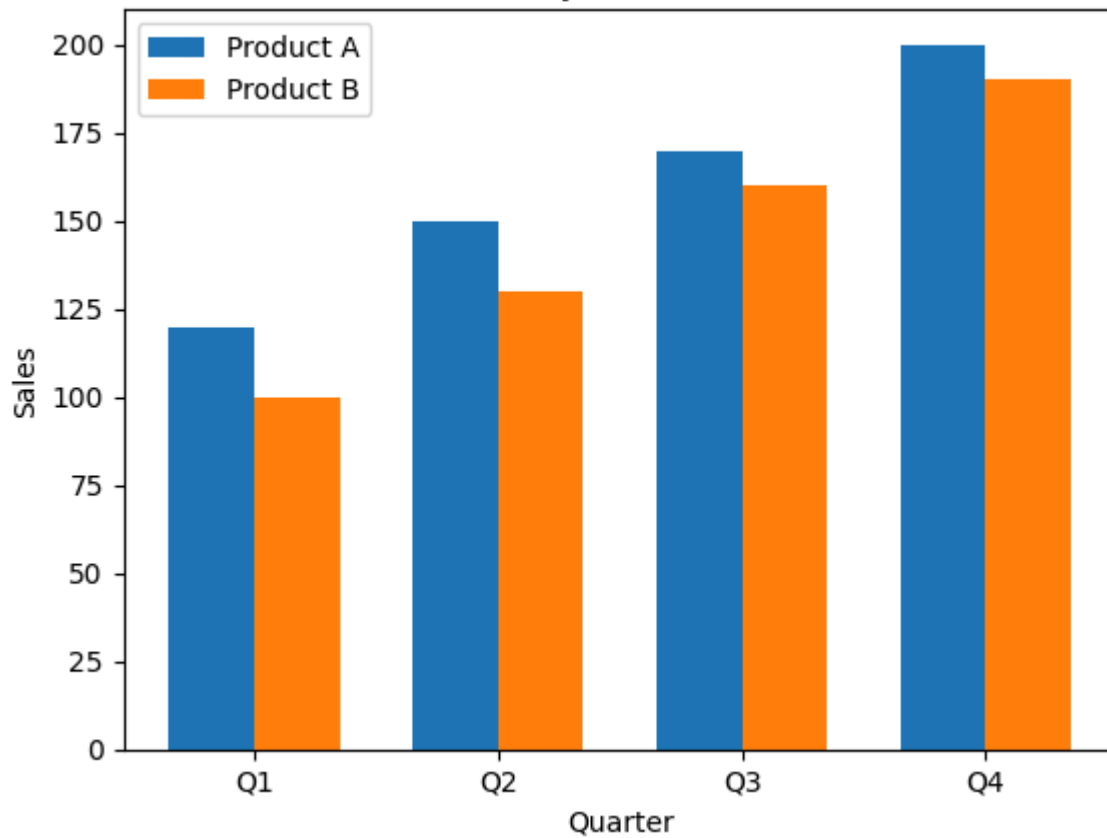
plt.xticks(x, quarters)

plt.title("Quarterly Product Sales")
plt.xlabel("Quarter")
plt.ylabel("Sales")

plt.legend()

plt.show()
```

Quarterly Product Sales



In [11]:

```
import matplotlib.pyplot as plt

departments = ["CS", "IT", "Business", "Arts"]

male = [60, 50, 40, 30]
female = [40, 45, 55, 70]

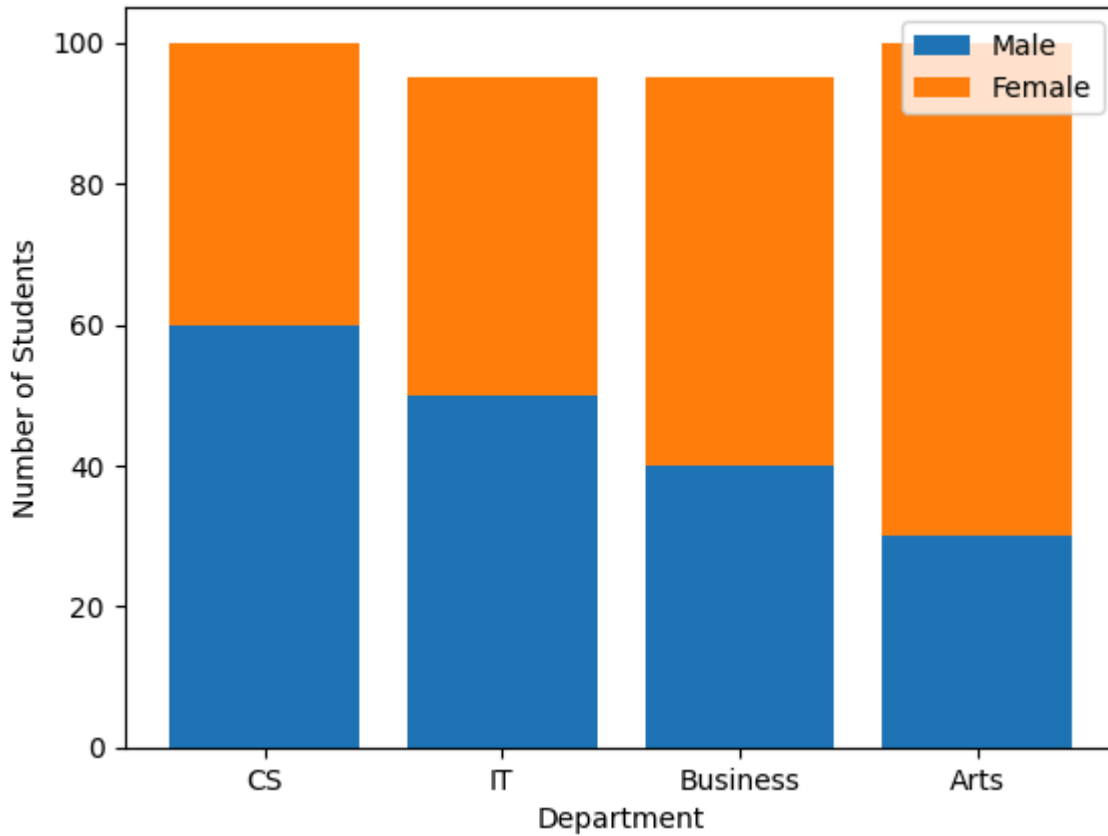
plt.bar(departments, male, label="Male")
plt.bar(departments, female, bottom=male, label="Female")

plt.title("Students by Department")
plt.xlabel("Department")
plt.ylabel("Number of Students")

plt.legend()

plt.show()
```

Students by Department



In [12]:

```
import matplotlib.pyplot as plt

months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]

product_A = [100, 120, 140, 160, 180, 210]
product_B = [90, 110, 150, 145, 170, 190]

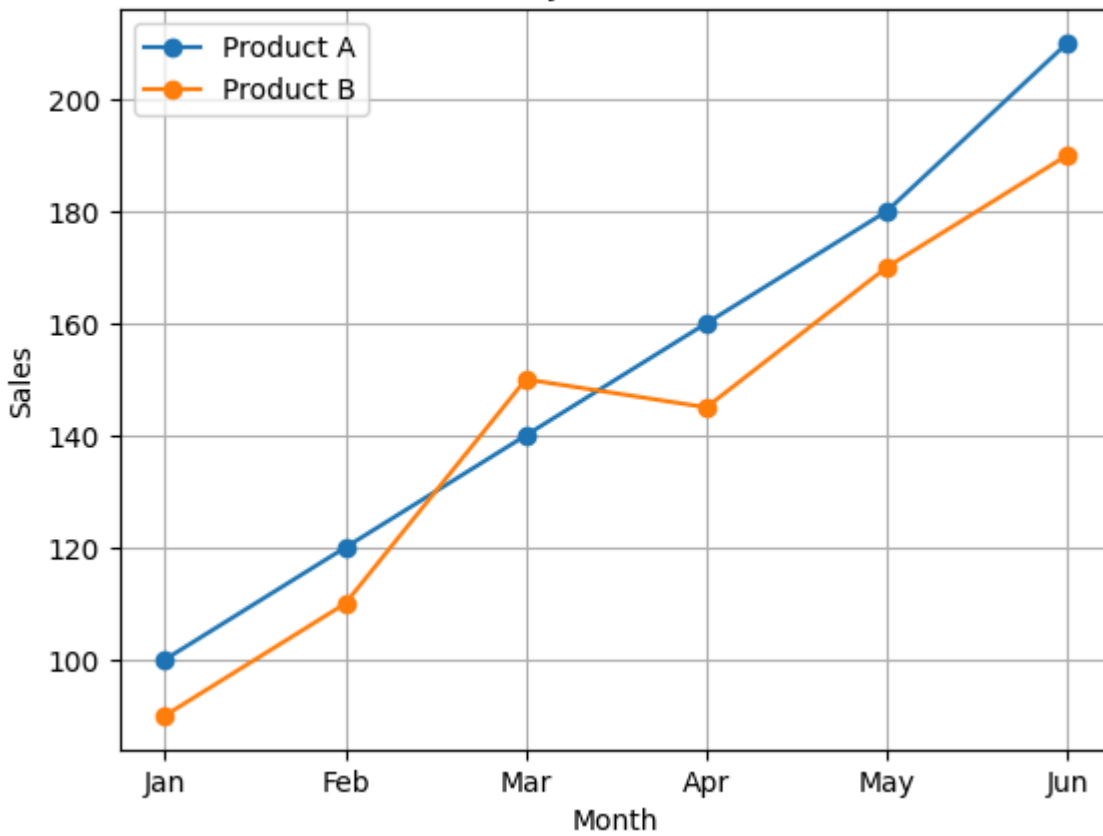
plt.plot(months, product_A, marker="o", label="Product A")
plt.plot(months, product_B, marker="o", label="Product B")

plt.title("Monthly Product Sales")
plt.xlabel("Month")
plt.ylabel("Sales")

plt.legend()
plt.grid()

plt.show()
```

Monthly Product Sales



In [13]:

```
import matplotlib.pyplot as plt
import numpy as np

data = np.array([
    [20, 40, 60],
    [25, 45, 65],
    [30, 50, 70],
    [35, 55, 75],
    [40, 70, 100]
])

days = ["Mon", "Tue", "Wed", "Thu", "Fri"]
times = ["Morning", "Afternoon", "Evening"]

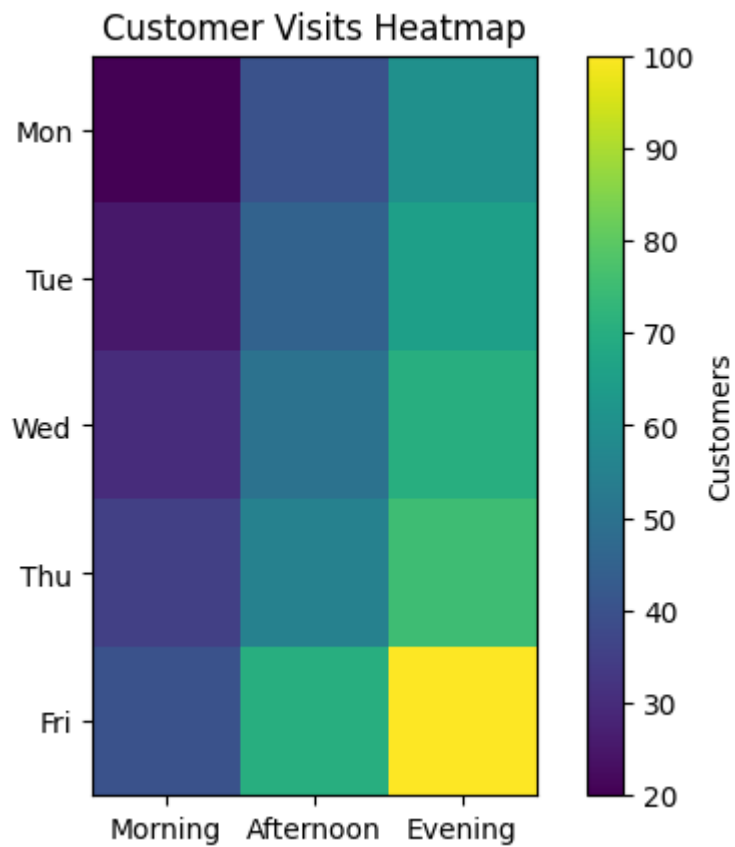
plt.imshow(data)

plt.xticks(range(len(times)), times)
plt.yticks(range(len(days)), days)

plt.colorbar(label="Customers")

plt.title("Customer Visits Heatmap")

plt.show()
```



In [14]:

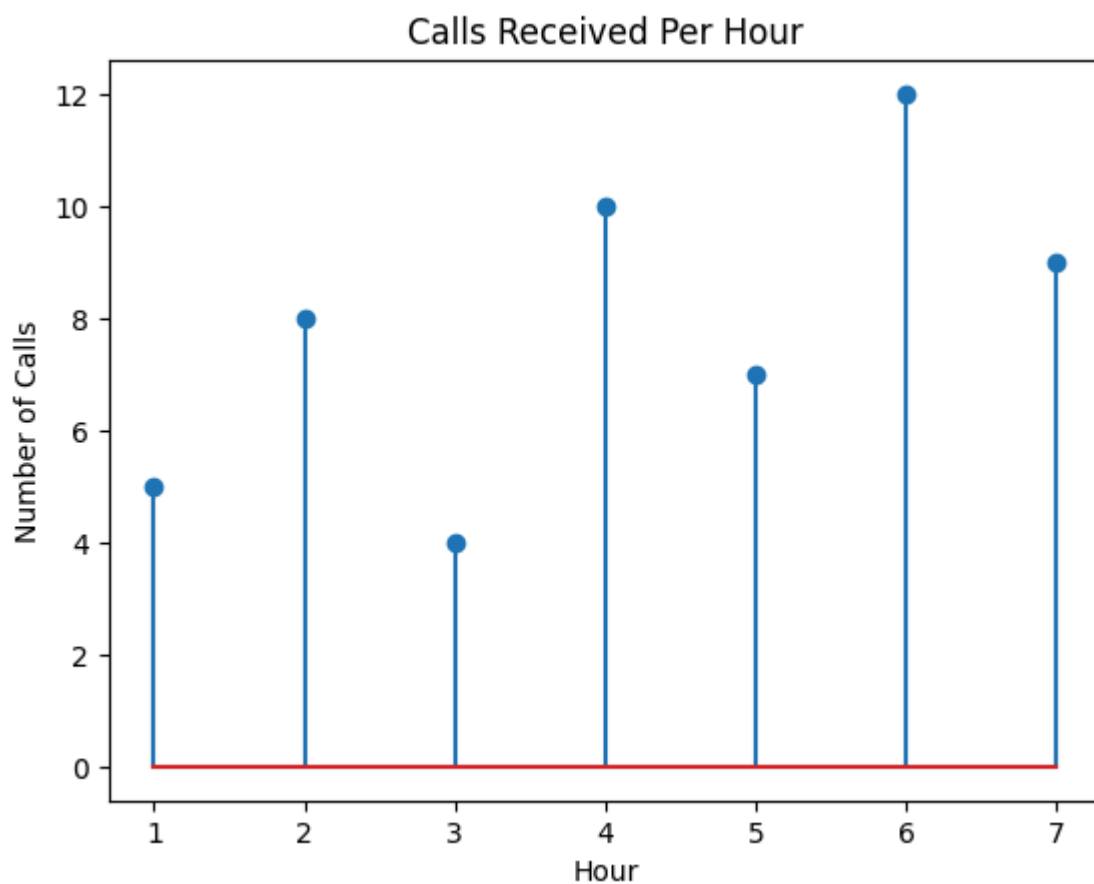
```
import matplotlib.pyplot as plt

hours = [1, 2, 3, 4, 5, 6, 7]
calls = [5, 8, 4, 10, 7, 12, 9]

plt.stem(hours, calls)

plt.title("Calls Received Per Hour")
plt.xlabel("Hour")
plt.ylabel("Number of Calls")

plt.show()
```



In [16]:

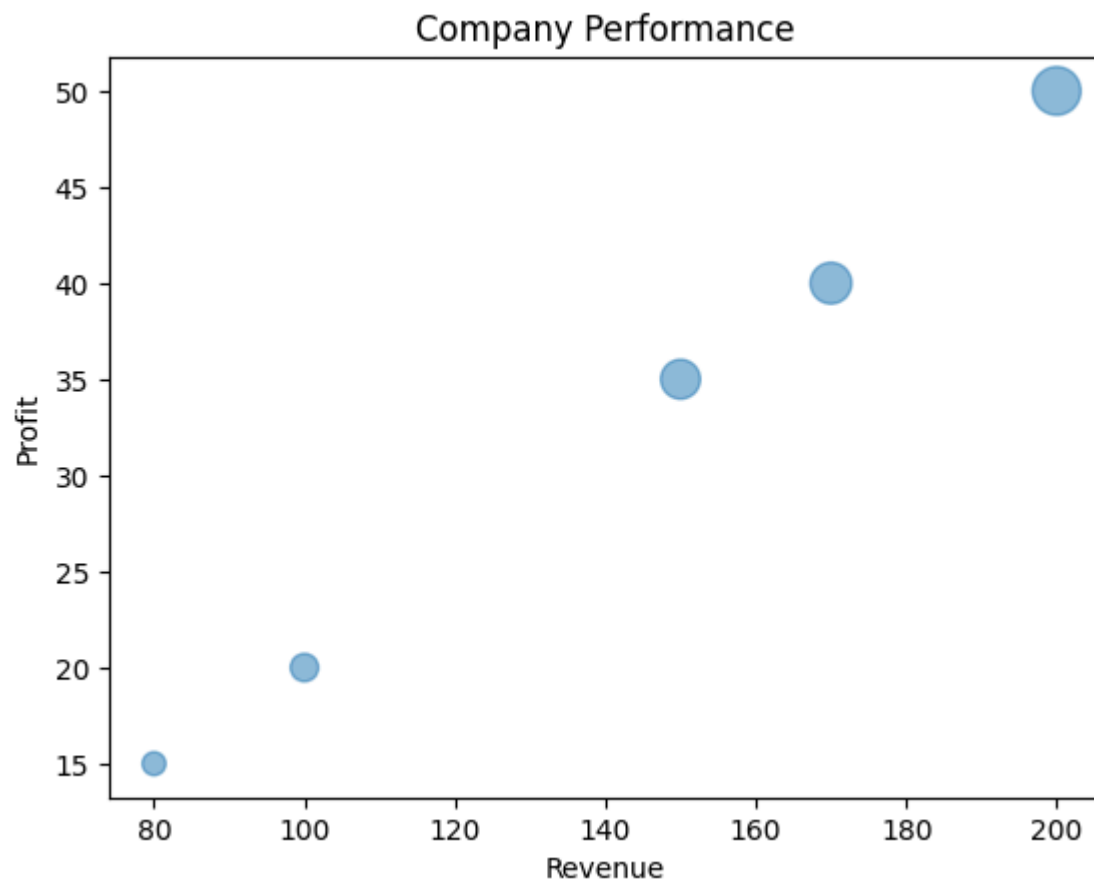
```
import matplotlib.pyplot as plt

revenue = [100, 150, 80, 200, 170]
profit = [20, 35, 15, 50, 40]
employees = [100, 200, 70, 300, 220]

plt.scatter(
    revenue,
    profit,
    s=employees,
    alpha=0.5
)

plt.title("Company Performance")
plt.xlabel("Revenue")
plt.ylabel("Profit")

plt.show()
```



In [17]:

```
import matplotlib.pyplot as plt

experiments = ["A", "B", "C", "D"]

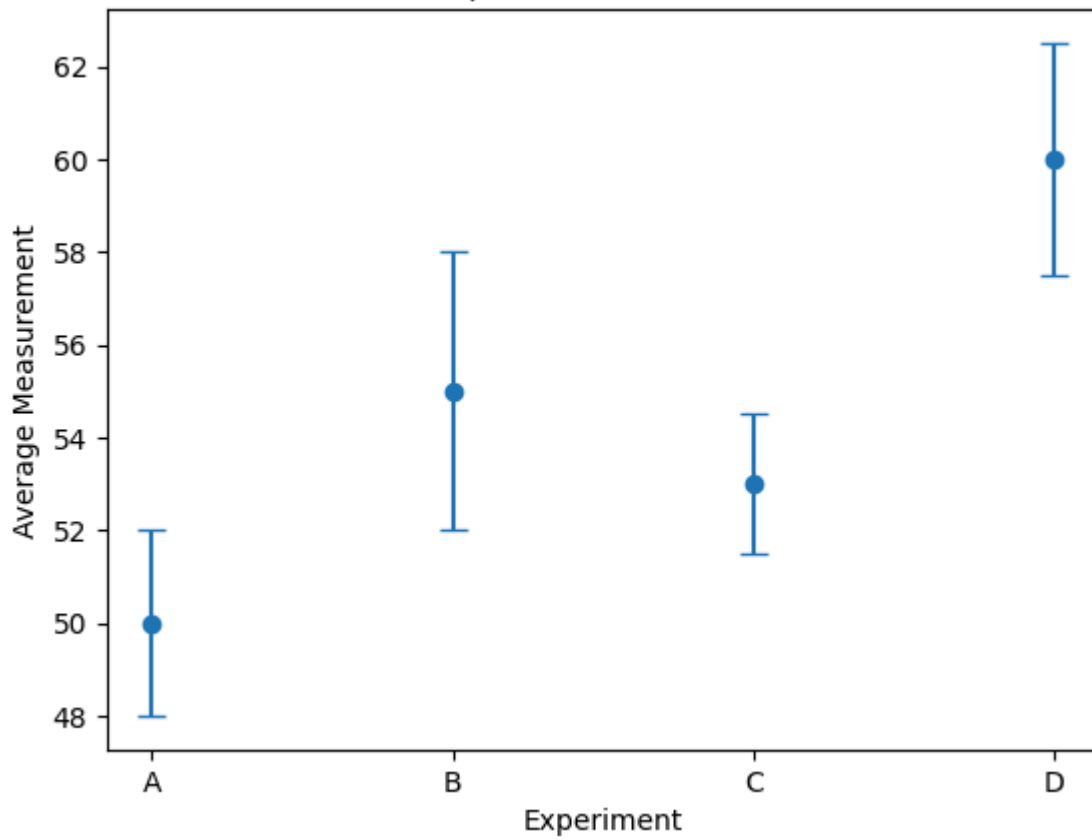
average = [50, 55, 53, 60]
error = [2, 3, 1.5, 2.5]

plt.errorbar(
    experiments,
    average,
    yerr=error,
    fmt="o",
    capsize=5
)

plt.title("Experimental Results")
plt.xlabel("Experiment")
plt.ylabel("Average Measurement")

plt.show()
```

Experimental Results



In [18]:

```
data = {  
    "Name": ["Amit", "Sara", "John", "Neha", "Ali"],  
    "Age": [20, 21, 19, 22, 20],  
    "Study_Hours": [4, 7, 3, 8, 5],  
    "Marks": [65, 88, 55, 92, 72]  
}
```